MILITARY SPECIFICATION SHEET

DIAL, CONTROL, MULTITURN COUNTERS, VERNIER, .88 DIAMETER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the dial described herein shall consist of this specification and the latest issue of MIL-D-28728.

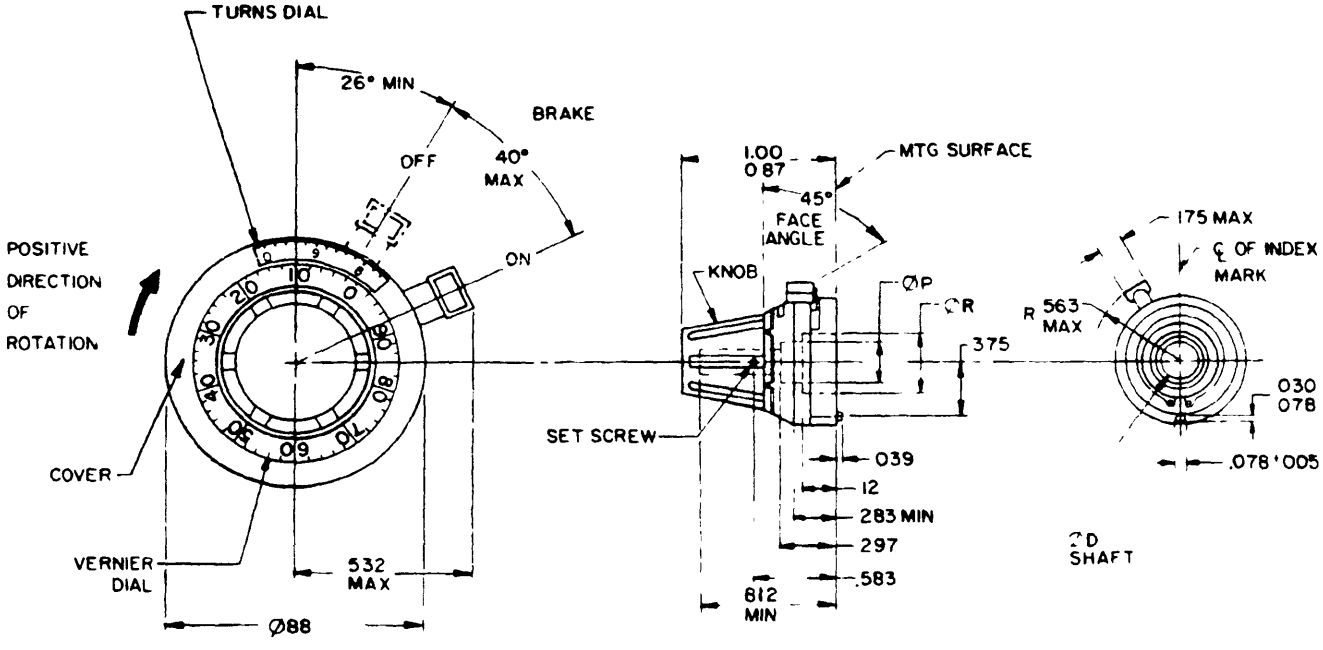


FIGURE 1. DIAL, CONTROL, MULTITURN COUNTERS, VERNIER, .88 DIAMETER

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:52Z
Check the source to verify that this is the current version before use.

TABLE I. DASH NUMBERS AND DESCRIPTION

DESCRIPTION							
DASH NO	GEAR RATIO CODE XX.X:1 (SEE NOTE 5)	COLOR CODE 1/			$\varnothing D$ +.002 -.000	$\varnothing P$ +.010 - 000	R MIN
		KNOB BRAKE LEVER	COVER	DIALS			
001	10.0	B	C	C	.252	.375	.51
002	10.0	B	C	B	.252	.375	.51
003	15.0	B	C	B	.127	.250	.37
004	15.0	G	C	C	.252	.375	.51

1/ B = Black; C = Satin chrome; G = Gray;

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. All dimensions are in inches.

2. Unless otherwise specified, tolerances are $\pm$.010 for three place decimals, $\pm$.02 for two place decimals and $\pm 5^{\circ}$-0' on angles.

3. Numerals: .050 high and graduations are indented and color filled. White on black dial and black on chrome dial.

4. Indicia:
   Vernier scale - graduations every 36° with subdivisions every 7.2°. Numerals from 0 to 90 in tens, marked every other graduation.
   Turns scale - numerals with optional graduations to indicate number of turns. Three numerals must be visible in the window at all times.

5. Gear ratio: The first two digits of the gear ratio code indicate whole number of turns of the knob needed to rotate the shaft one full turn (see Table I).

REQUIREMENTS:

1. Material: All components shall be of suitable materials to meet requirements.

2. Finish (see Table I):
   Color in accordance with the following:
   Chrome satin, natural in accordance with QQ-C-320, Class 1, Type II.
   Black Lusterless Paint, color number 37078 in accordance with FED-STD-595.
   Gray Lusterless Paint, color number 36231 in accordance with FED-STD-595.
   White Lusterless Paint, color number 37875 in accordance with FED-STD-595.

3. Operating temperature range: -65° F to 160° F.

4. Weight: 0.6 ounces maximum.

5. Torque: Operational - 1.5 inch ounce $\pm$15%.
   Locking brake - 25 inch ounce minimum

6. Method of count: Continuous rotation of vernier scale clockwise for increase 0 to 99 to 0 in count and counter-clockwise for decrease in count. One complete turn of knob and vernier scale shall move turns dial one increment.

7. Readout accuracy: 0.5 percent of full count (1 part in 500).

8. Life: 1,000,000 cycles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:52Z
Check the source to verify that this is the current version before use.

9. Operational speed: 60 RPM maximum.

10. Part numbers: The part number consists of the letter "M", the basic number of this specification sheet and a dash number taken from Table I.

Example: M28728/8-001 indicates - Dial, ten turn counter, vernier .88 diameter, black knob, satin chrome cover and dial, with .250 diameter shaft hole.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AV, MI
Navy - AS, OS, SH
Air Force - 80
DLA - IS

User activities:
Army - AR, ME
Air Force - 17

Preparing activity:
Navy - EC

Agent:
DLA - IS

(Project 5355-0191)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:52Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

SPACE AND NAVAL WARFARE SYSTEMS COMMAND
WASHINGTON D C 20363-5100

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
SPACE AND NAVAL WARFARE SYSTEMS COMMAND
ATTN: SPAWAR 003-121
WASHINGTON, DC 20363-5100

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:52Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-28728/8B Rev B	DIAL, CONTROL, MULTI-TURN COUNTER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T13:52Z
Check the source to verify that this is the current version before use.